

Fitting Private Equity into the Total Portfolio Framework

ALEXANDER RUDIN, JASON MAO, NAN R. ZHANG,
AND ANNE-MARIE FINK

ALEXANDER RUDIN
is managing director
of State Street Global
Advisors in Boston, MA.
alexander_rudin@ssga.com

JASON MAO
is vice president of State
Street Global Exchange
in Boston, MA.
xmao@statestreet.com

NAN R. ZHANG
is vice president of State
Street Global Exchange
in Cambridge, MA.
nzhang2@statestreet.com

ANNE-MARIE FINK
is managing director
of State Street Global
Advisors in Stamford, CT.
anne-marie_fink@ssga.com

KEY FINDINGS

- By applying a somewhat novel econometric technique to a proprietary dataset, we offer a way to separately estimate systematic and idiosyncratic risks of private equity (PE) programs. Our methodology controls for data overfitting and program concentration.
- Observed volatility and equity beta of PE are much lower than that of public equity. We believe this is not an aberration but instead a fundamental consequence of PE being less exposed to the excess volatility of public equity markets.
- With the exception of the tumultuous period of the global financial crisis (GFC), risk properties of private equity have been surprisingly persistent; if anything, systematic risks of private equities decreased slightly after the GFC.

ABSTRACT: *In this article, the authors propose a risk estimation model that addresses both smoothness and idiosyncratic risk dynamics of narrow private equity portfolio returns. The authors subsequently apply the model to a broad set of private equity return streams with tightly controlled diversification properties. They show that increased model complexity is detrimental to the model's forecasting power and that idiosyncratic risks increase as one goes between hypothetical investment in a broad (albeit noninvestable) private equity index and a real-life, narrowly diversified portfolio of private equity funds. The authors also find that private equity returns have demonstrated a level of exposure to public equity markets that may be considered surprisingly low and offer a possible qualitative justification for this phenomenon. Finally, the authors incorporate the newly obtained private equity risk model into a general portfolio construction framework and demonstrate how to study the inevitable trade-offs between the costs and benefits of increased portfolio diversification.*

TOPICS: *Portfolio theory, portfolio construction, equity portfolio management**

Private equity has long been a core part of many institutional investor portfolios despite the challenges of fitting this asset class into traditional asset allocation frameworks. Most such frameworks use an asset's return series as the basis for analyzing allocations. This approach has complications in the case of private equity. First, the private equity time series itself is smoothed owing to the difficulty in obtaining mark-to-market valuations for private companies and the broadly observed lag private company valuations tend to demonstrate relative to their public market counterparts. The second complication is the inability to invest directly in any of the available private equity indexes.

*All articles are now categorized by topics and subtopics. **View at PM-Research.com.**

In this article we propose a risk estimation process that addresses both difficulties. Our proposed way to handle smooth private equity returns is an extension of work by Pedersen, Page, and He (2014) and a much earlier work by Conner (2003), to which we make a modest but helpful methodological improvement. Our main contribution is that we explicitly study the risk gap between hypothetical investment in a broad (albeit noninvestable) private equity index and a real-life, narrowly diversified portfolio of private equity funds. We measure (empirically) how the gap widens as we reduce number of investments in such portfolios. We study the robustness of factor-based risk forecasting, and we make recommendations for the econometric model for private equity. We subsequently translate our findings into a framework for allocating to private equity within a total portfolio and focus on the trade-offs between diversification benefits and costs.

Before proceeding further, however, we cannot sidestep a general question on whether it is fruitful to apply *any* econometric technique to private equity returns given the perceived opaque nature of private equity valuations and whether accounting changes between 2006 and 2008, first in Europe and then in the United States, changed the way we should approach this subject.¹ Indeed, prior to 2006, private equity funds could, at their discretion, keep their investments at cost in terms of valuation. European funds converted to fair value accounting in 2006 with the adoption of IAS 39 (in 2005), and US funds soon followed suit with the adoption of ASC 820 (formerly known as FSAS 157) in 2008.

A fair amount of research has been conducted to determine whether private equity fund net asset values (NAVs)—and hence observed returns—do indeed contain useful information and, if so, whether the amount and quality of this information has improved since accounting changes took place. Jenkinson et al. (2016) found that although NAVs were associated with fair values even prior to the adoption of fair value measurement standards in 2008, NAV accuracy improved even further after these standards were adopted, especially for the buyout funds. Crain and Law (2017) found that

use of fair value accounting reduces valuation bias and increases valuation accuracy. Brown, Gredil, and Kaplan (2018) found reported NAVs to have little systematic bias in most cases. Easton, Larocque, and Stevens (2018) focused exclusively on the US-based funds and observed that private equity fund NAVs predict ex post future net distributions to fund investors more accurately after accounting rules changed in 2008. The special nature of the 2006–2008 period was studied in an interesting work by Welch and Stubben (2018). The authors focused on European funds during this period precisely because these were the years when European funds had switched to fair value accounting but US funds had not yet done so, providing an interesting control group. The authors were able to isolate the effects of overall market gyrations (which happened to be huge because of the financial crisis) and accounting changes in Europe during this period.

In all, researchers have concluded (with various caveats, of course) that despite subjectivity in valuing private companies, private equity fund returns do indeed contain useful information, and the accounting changes of 2006–2008 only improved the quality of this information. This makes techniques like the one we suggest in our work applicable to past data and even more applicable going forward.

Before describing our findings, we would like to also point out what this article is *not* trying to accomplish. There is already a significant body of work that aims to understand systematic drivers (factor exposures) of alternative investment strategies. Pedersen, Page, and He (2014) conducted a good literature review of the subject and provide a detailed empirical analysis of how various factors affect private equity, real estate, and other alternatives. It was not our goal to discover the best factor representation for private equity beyond an intuitive exposure to public equity market returns.

Although some of the concepts introduced in this article are somewhat transferrable to other alternative asset classes, it was also not our goal to analyze alternatives other than private equity. We were fortunate to have access to a robust dataset for private equity funds. We did not have access to a similar quality data for real estate or other alternatives and thus could not perform analogous exercises with the same rigor.

We would also like to differentiate our approach to studying private equity from the public market

¹ We are grateful to the anonymous reviewer for pointing out the accounting dimension of the problem; our study into the variation of risk estimates across time is a direct result of the reviewer's comments.

equivalents (PME) analysis used in highly publicized works by L'Her et al. (2016) and Phalippou (2014). These authors concluded that, at least in the case of the buyout funds, there is “no significant outperformance of buyout fund investments versus their public market equivalents on a dollar-weighted basis” after the public market proxy is appropriately risk adjusted. Although insightful, these conclusions, by the nature of PME analysis, apply to extremely long-term (10+ years) time horizons. The time-varying returns framework employed by us and by earlier authors such as Pedersen, Page, and He (2014) and Conner (2003) aim to improve risk estimation and portfolio construction over the medium-term investment horizon adopted by most institutions. When considering these applications, PME-based conclusions made by L'Her et al. (2016) and Phalippou (2014) are of somewhat limited value. We will provide additional insights on why our results and results by L'Her et al. (2016) and Phalippou (2014) are not mutually exclusive when we make our concluding remarks.

Another paper we would like to mention is that by Schmidt (2004), which is related to our present findings because in it Schmidt used a variation of a bootstrapping simulation to imitate private equity performance while maintaining some control over diversification. Schmidt also aimed to build a framework for determining optimal allocation to this asset class within the total portfolio. Our work is substantially different. The simulation process by Schmidt (2004) created matches between private companies and similar public ones to imitate private equity portfolios, whereas we work directly with private equity funds. Schmidt used what effectively was a lifetime PME approach to measuring private equity returns, whereas we study rolling returns that are easy to compare and contrast with asset classes other than private equity. We also establish a return unsmoothing methodology and create a framework for measuring factor exposures within private equity, something Schmidt (2004) did not do and to which his framework is not well suited.

Finally, we do not prefer a particular variation of the portfolio construction methodology over all others. Most portfolio construction techniques require estimations of future values of volatility and correlation across portfolio assets; a process of developing those estimates is where our contribution resides.

RISK ESTIMATION THROUGH UNSMOOTHING

Academic and practitioner communities alike broadly accept that many alternative asset classes exhibit stale pricing patterns or smooth returns. In the particular case of private equity, this phenomenon has been noted by many researchers (see, for example, Anson 2002) and is linked to the market practice of obtaining valuations of private companies through self-reporting and/or appraisals. There is a broad body of literature on the subject, from which we would like to highlight two contributions that are particularly relevant to this section of our article. Conner (2003) noted that directly using private equity or other “smooth return” time series is misleading because the smoothness hides the volatility of the underlying “true economic process” and obscures relationships with other liquid assets in the portfolio. Conner (2003) suggested using a methodology from the real estate world (Geltner 1991) that estimates actual volatility by unsmoothing the time series of returns.

Pedersen, Page, and He (2014) went one step further and noted that in addition to needing to be unsmoothed, reported return time series ought to be analyzed through the lens of relevant factor exposures. Such analysis seeks to separate alternative investment risks that can be attributed to systemic factors from the purely idiosyncratic risk component. The results of such analysis are highly relevant to the forward-looking risk estimation process and ultimately for portfolio construction.

In the remainder of this section we build an econometric model that is philosophically similar to ones used by Pedersen, Page, and He (2014) and Conner (2003), with a change designed to improve the robustness of the parameter estimation process. We later use this model to estimate the risk properties of what we call *investable mini-programs* in the private equity space.

Let us denote the observed smooth return of a private equity asset as r_t^O . The unsmoothing approach starts by recognizing the link between this return time series and the one for an unobserved true economic process return r_t^E . Both Conner (2003) and Pedersen, Page, and He (2014) linked the two in the following way:

$$r_t^O = \sum_{j=0}^Q w_j r_{t-j}^E \quad (1)$$

EXHIBIT 1

Regression Analysis Results for Private Equity Buyouts

- Index: SSPE Equal-Commitment BO (2000–2014)
- Factor: S&P 500

Max # of Lags (Q)	Alpha	Beta to S&P	Volatility	% Explained (in sample)	
No Lags	9.23%	0.37	9.5%	44%	
1 Lag	4.64%	0.53	13.3%	46%	} For “Unsmoothed” Time Series
2 Lags	3.34%	0.71	16.6%	52%	
3 Lags	3.25%	0.72	16.8%	53%	
4 Lags	2.84%	0.85	19.4%	56%	

Here Q is the maximum number of lags, and weights w_j sum to one. Such a formulation, together with empirical data, allowed both authors to estimate weights w_j and subsequently create a time series for true returns and evaluate its risk properties.

Additionally, Pedersen, Page, and He (2014) noted that the true economic process may be expressed as a linear combination of factor exposures denoted as F_t^i , in addition to some idiosyncratic risks:

$$r_t^E = \alpha^E + \sum_{i=1}^N \beta_i F_t^i + \varepsilon_t^E \quad (2)$$

Substituting Equation 2 into Equation 1, Pedersen, Page, and He (2014) obtained an econometric model for factor exposures in the presence of smooth returns, completing the risk estimation process.

Although an extremely helpful breakthrough, this econometric model carried two drawbacks that we would like to address before proceeding to the rest of our analysis. First, Pedersen, Page, and He chose to estimate smoothness parameters w_j and factor betas β_i in two consecutive steps. Because each estimation involves a regression and introduces a layer of estimation errors, there is a benefit in estimating all parameters simultaneously in a single regression. Second, Pedersen, Page, and He’s process created a regression equation that carries an autocorrelated error term, requiring an additional step of involving a Newey–West procedure to correct for it.

We would like to point out that both drawbacks can be addressed if we start from an alternative but equivalent version of Equation 1 (see also Stefek and Suryanarayanan 2012, who applied a similar approach to real estate):

$$r_t^O = \theta_0 r_t^E + \sum_{j=1}^Q \theta_j r_{t-j}^O \quad (3)$$

Here θ_j are unknown coefficients that can be estimated through regression analysis. Equation 3 is equivalent to Equation 1 in the sense that, knowing θ_j , one can calculate w_j and vice versa (also, $\sum_{j=0}^Q \theta_j = 1$). Equation 3 also has two material benefits. First, as long as one knows θ_j , true economic returns can be easily expressed through known observed returns by simply rearranging Equation 3. Second, substituting Equation 2 into Equation 3 yields a regression model with an error term that is not autocorrelated:

$$r_t^O = \alpha + \left(1 - \sum_{j=1}^Q \theta_j\right) \sum_{i=1}^N \beta_i F_t^i + \sum_{j=1}^Q \theta_j r_{t-j}^O + \varepsilon_t \quad (4)$$

Despite its seeming complexity, Equation 4 allows, through a straightforward regression analysis, simultaneous estimation of smoothing parameters θ_j and factor betas β_i .

Exhibit 1 illustrates the results of this approach applied to private equity buyouts represented by the State Street Private Equity (SSPE) index.² In an attempt to keep the calculation simple, we chose to use only one factor—the S&P 500 total return index—to represent a well-established connection between private and public equity markets. For a more complete picture of factor exposures, please see Pedersen, Page, and He (2014).

²We chose the State Street Private Equity index for this analysis to ensure consistency with the rest of our article. Applying similar analysis to other popular private equity indexes such as Burgiss or Cambridge Associates yields results that are essentially identical.

We also, for now, use the full time period for which data are available to us (i.e., between 2000 and 2014). For purposes that will soon be clear, we included results for a range of models, each corresponding to a different maximum number of lags Q .

These results are generally in line with findings by Pedersen, Page, and He (2014). Switching from a reported smooth time series to a true economic one has the general effect of increasing both volatility and factor exposure estimates for alternative assets with smooth observed valuations. We note that half or more of the risk in those returns (after complete unsmoothing) can be attributed to public equity proxies, with the rest complementary to public equities³ and carrying a fair amount of alpha.

Although this technique is insightful, several challenges remain. The first and most obvious one is the need to settle on a number of lags Q to use in the econometric model of Equation 4. Some researchers recommend using some sort of information criterion to choose (see, e.g., Liew 2004). This method tends to recommend a large number of lags, attempting to absorb as much information as possible into the regression. Others go for a more parsimonious choice of a single lag or use their perceived knowledge of how stale pricing actually works.⁴ The consequences of the choice are significant, as evidenced by Exhibit 1.

The second challenge is that a factor and smoothness analysis of Equation 4 is typically applied to a broad private equity index containing thousands of individual fund contributions. A typical institutional private equity program encompasses dozens, and in most cases less than 100, individual fund investments. It is thus much narrower in scope than any industry-wide, uninvestable private equity index. It is not immediately obvious whether the factor model built for a real-world, executable program demands

a different parametrization than one built off index data. Intuitively, it seems likely that a narrower program will carry higher idiosyncratic risks and thus higher volatility of the residual term of Equation 4.

We believe both challenges can be addressed simultaneously and consistently if we apply our model to the historical returns of a large number of long-running private equity programs instead of applying it to a single index. This insight underlies our research contribution: By controlling the diversification level of those programs, we could measure how idiosyncratic errors depend on the level of diversification. At the same time, testing how well our models predict private equity returns can help us choose the optimal value for the number of lags Q to use and make some general observations on the efficacy of our approach.

There is no readily available private equity program dataset to use for such analysis; it simply does not exist in the industry. Instead, we set out to proxy such a dataset by combining anonymized historical returns of private equity partnerships available to us through State Street Global Exchange.

SIMULATED MINI-PROGRAMS AND OPTIMAL FACTOR MODELS FOR PRIVATE EQUITY

Derived from actual cash flow data of State Street's limited partner clients who make commitments to private equity funds, SSPE is one of the most detailed and accurate private equity datasets in the industry today. Because this dataset does not depend on voluntary reporting of information, it is less exposed to biases common among other industry indexes, resulting in more reliable and consistent content. As of Q1 2019, SSPE comprises more than 3,000 funds globally, representing more than \$3.0 trillion in capital commitments, with cash flow data back to 1980. Studying such a dataset *on an anonymized basis* provides an opportunity to gain analytical insight into an otherwise opaque asset class.

Our first step was to construct a number of paths, each corresponding to the historical performance of an artificially constructed mini-program in a particular private equity strategy. Each path was constructed under assumptions that replicate typical investor behavior. Every vintage year, the same size commitments were made to the strategy (say, \$1). This commitment was split equally to a predefined number of funds within

³ Aside from the stale pricing for which an econometric approach can adjust, there are many reasons for variability in private equity returns above and beyond public equity gyrations. For instance, private equity managers time their allocations to private companies, whereas public indexes are fully invested at all times. Private equity managers are opportunistic and vary in their allocations across industries and geographic locations. Capital flows have different cadences in public and private markets. These variations are only a few examples.

⁴ For example, in the real estate literature it is common to choose four quarterly lags to accommodate the annual appraisal cycle in the United States.

the vintage. This number was kept constant through the years for each synthetic mini-program, allowing us to control the level of diversification. Funds were chosen randomly and without repetition.

In all, we created 500 such mini-programs (with calculated track records for the 15-year period between 2000 and 2014) for each level of program diversification, separately for buyout and venture capital funds. Diversification levels ranged from 1 fund per year to 10 funds per year. We also constructed an index for each strategy that corresponded to the performance of a hypothetical private equity program that is built along the same equal-commitment lines as the mini-programs but encompasses all available funds.⁵ We then analyzed both sets from a variety of perspectives. In our first set of numerical experiments, we covered what we considered the full sample for our analysis: 2000–2014. We chose to exclude the last five years (2015–2019) from the analysis to make sure we only include vintages that are sufficiently seasoned. We also describe results for other sample choices (pre-2006, 2006–2008, post-2008) in the following sections.

Determining the Optimal Number of Lags

Our first step was to build Equation 4–type models for the equal commitment indexes, by strategy, for Q ranging between 0 (no unsmoothing) and 4. We then set out to test the relative quality of these models by applying obtained parameters (θ_j^{INDEX} and β_i^{INDEX}) to the simulated mini-program paths and measuring how much of the performance of such mini-programs is explained by the model. Specifically, we measured the ratio of the variance of returns unexplained by the model to that of the total returns of the mini-programs.

Result are shown in Exhibit 2. In what we consider an important result, using complex models with a number of lags in excess of two is counterproductive: Instead of explaining away a larger portion of returns,

⁵ The performance of such indexes will rhyme with, but will not be identical to, that of better-known private equity indexes (including the State Street ones) because their construction is based on capitalization weights as opposed to equal weights. Our research shows that the largest 10% of funds by size counts for 34% of all capital raised in venture capital and a whopping 47% in the buyout. In other words, broad index performance to some degree reflects the peculiarities of the largest mega-funds, which tend to be more leveraged and carry higher beta exposure to the public equity markets.

such models introduce new risks (ratio greater than one). This outcome is not entirely a surprise. The data available to us for each index are quite limited; 15 years of data only produce 60 quarterly return observations. Exceedingly complex multifactor models applied to small datasets tend to overfit the data, leading to poor predictive power.

The good news is that using a one-lag, index-based model is helpful in explaining up to 50% of risk for a broad range of private equity portfolios (mini-programs) that we constructed in our tests.⁶

Idiosyncratic Risks and Mini-Program Diversification

This same approach allows us to study the idiosyncratic, hidden risks of investing in real-life, narrow private equity programs. Naturally, we used what we now know is the most robust, one-lag model for both buyout and venture capital strategies.

As in the previous section, we start by performing the regression of Equation 4 on the stream of private equity index returns with the purpose of estimating the unsmoothing constant θ and exposure to the equity factor β . Results are shown in Exhibit 3. The intercept of this regression demonstrates how much alpha private equity strategies add to beta-adjusted equity returns after smoothing is taken into account.

⁶ We also attempted to improve on the index-based model results by employing what we called a *shrinkage* approach, borrowing the methodology and terminology from the portfolio optimization literature (see, e.g., Brandt 2010). We built factor models separately for each asset (in our case, private equity mini-program) and then used average values for obtained betas and thetas as definitions of the shrinkage model for the space.

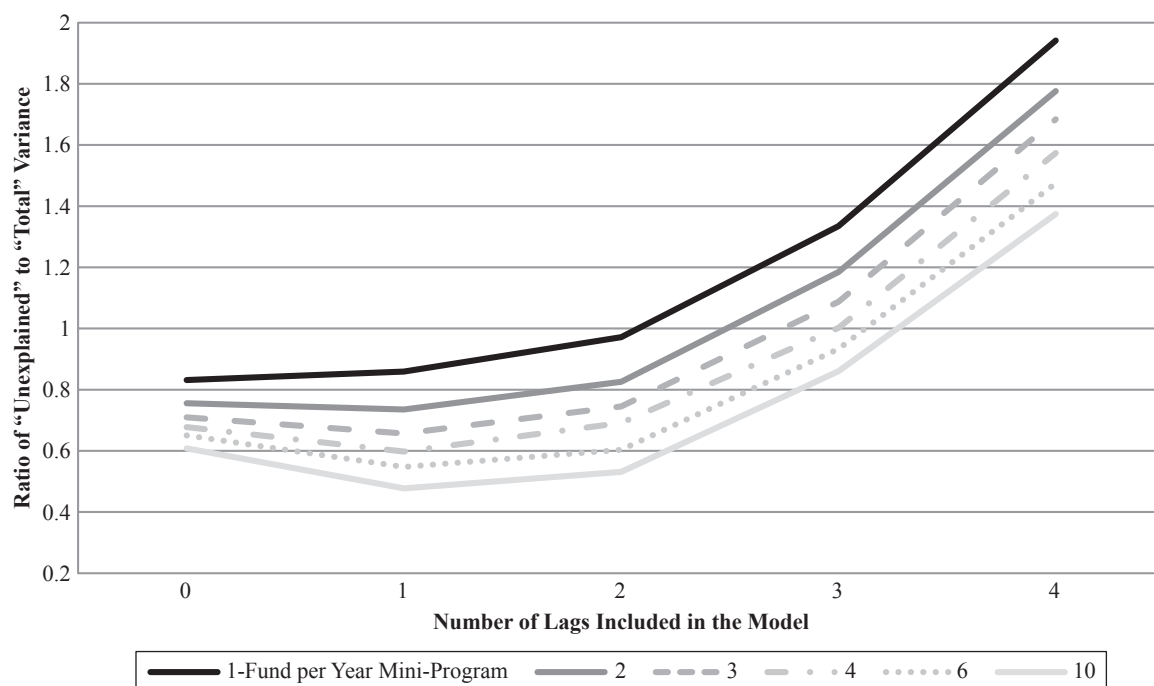
In our research, we split our 500 paths into two equal batches (250 each), built the shrinkage model for each strategy using the first batch, and then performed the same out-of-sample robustness testing of this model using data from the second batch. It appeared that the shrinkage model is more robust than the index-based model for high lags (three or more) but produces essentially the same results—slightly better—for small lags (zero to two). The single-lag model emerged as the most robust for the shrinkage model, just as it did for the index-based model for both venture capital and buyout strategies.

In all, our study showed that the shrinkage approach, although considerably more complex in terms of parameter estimation, added only limited value above and beyond the index-based model in terms of its predictive power. For the sake of simplicity, we use index-based parameter estimation for the remainder of this article.

EXHIBIT 2

Testing Model Robustness for Varying Number of Lags

Panel A: Testing Robustness of the Index-Based Model Out-of-Sample Buyout Strategy



Panel B: Testing Robustness of the Index-Based Model Out-of-Sample Venture Capital Strategy

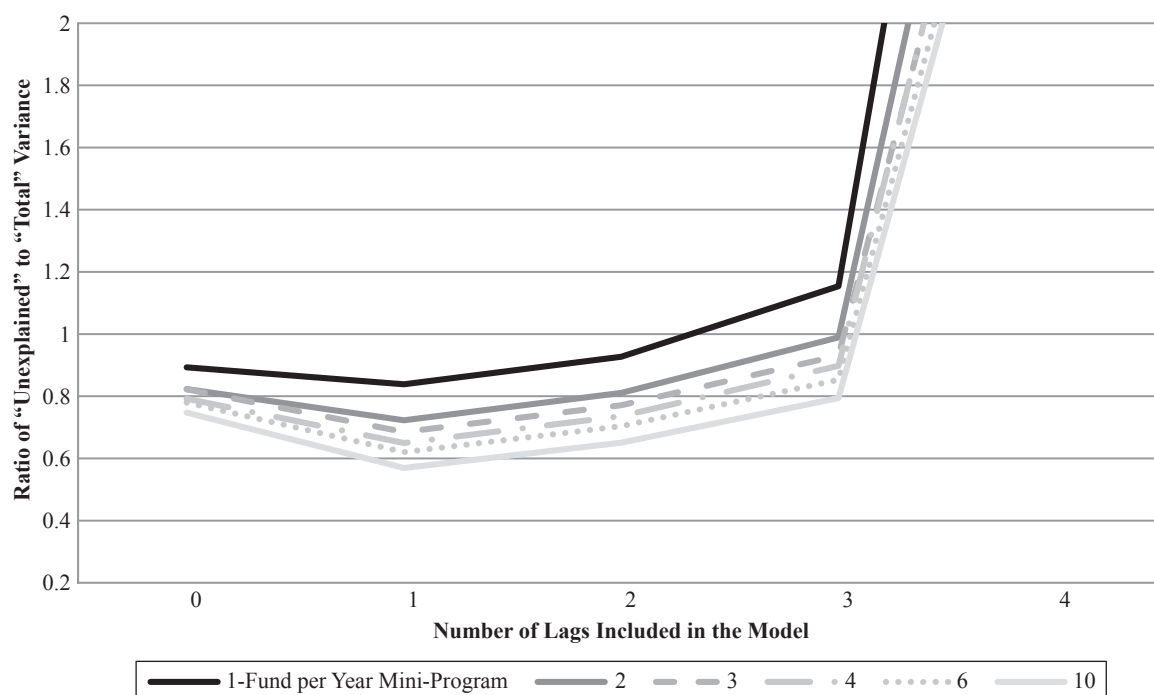


EXHIBIT 3

Index Model Summary (one lag): Data for 2000–2014

Panel A: Buyout Strategy

	α	β	θ	
No. of Funds per Year				Standard Deviation of the Residual
1	4.6%	0.53	0.39	15.6%
2				11.2%
3				9.4%
4				8.6%
5				8.3%
6				7.8%
7				7.4%
8				7.2%
9				7.0%
10				6.9%

Panel B: Venture Capital Strategy

	α	β	θ	
No. of Funds per Year				Standard Deviation of the Residual
1	1.0%	0.53	0.31	22.7%
2				17.7%
3				14.8%
4				12.9%
5				12.3%
6				11.5%
7				11.1%
8				10.5%
9				10.2%
10				9.8%

We then continue by subtracting returns obtained through such index model from each of the simulated mini-programs' return streams and measure the volatility of the obtained residual (*tracking error*). Exhibit 3 includes these results as well. Exhibit 4 shows the same results visually (now in terms of variance as opposed to standard deviation) while adding factor-based variance for completeness.

It is gratifying to see that for the reasonably diversified private equity programs, index-based analysis can explain (and hopefully forecast on a go-forward basis) up to 50% of the risk in terms of program variance. It is also painfully obvious that as the programs become narrower, with only few funds added per year, total risk rises dramatically while the percentage explained by the index model drops. Not surprisingly, idiosyncratic risks begin to overwhelm systemic risks in concentrated portfolios. The magnitude of this effect emphasizes once again the dangers of underdiversification in private equity.

EVOLUTION OF PRIVATE EQUITY RISKS ACROSS TIME

As we mentioned in the beginning, accounting treatment of private equity funds has evolved significantly over the years. European funds adopted fair value accounting in 2005 by approving IAS 39 and in

2006 converted to new accounting rules that may (or may not) have changed risk dynamic of private equity fund returns starting with 2006. In turn, the United States soon followed suit with approving of ASC 820 (formerly known as FSAS 157) in 2008, with changes actually adopted starting in 2009. This naturally splits our 2000–2014 sample into three parts: (1) 2000–2005, (2) 2006–2008, and (3) 2009–2014. We ran the same analysis separately for each of the three periods.

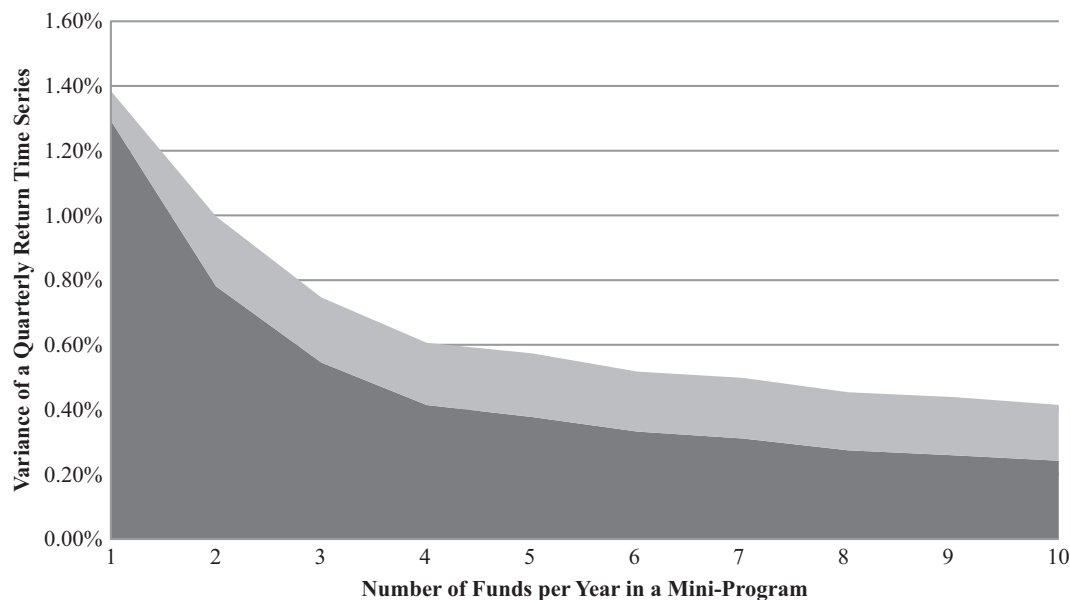
Many of the findings of the previous section remained essentially unchanged. A single-lag model is still the most robust of the possible alternatives, and all further results are produced using a single-lag model. The diversification dynamic when more concentrated mini-programs, on average, exhibit higher levels of idiosyncratic risks also remains intact, with standard deviations of residuals remaining close to the levels shown in Exhibit 3 for all three of the selected subperiods. Variability of risk characteristics was material only in factor exposure to equity markets and in the smoothing parameter. We summarize results in Exhibit 5, where, for convenience, we include the full-sample risk characteristics from Exhibit 3.

These results warrant several comments. First, estimated equity beta figures before (<2006) and after (>2008) accounting changes are surprisingly close to each other and surprisingly moderate in magnitude.

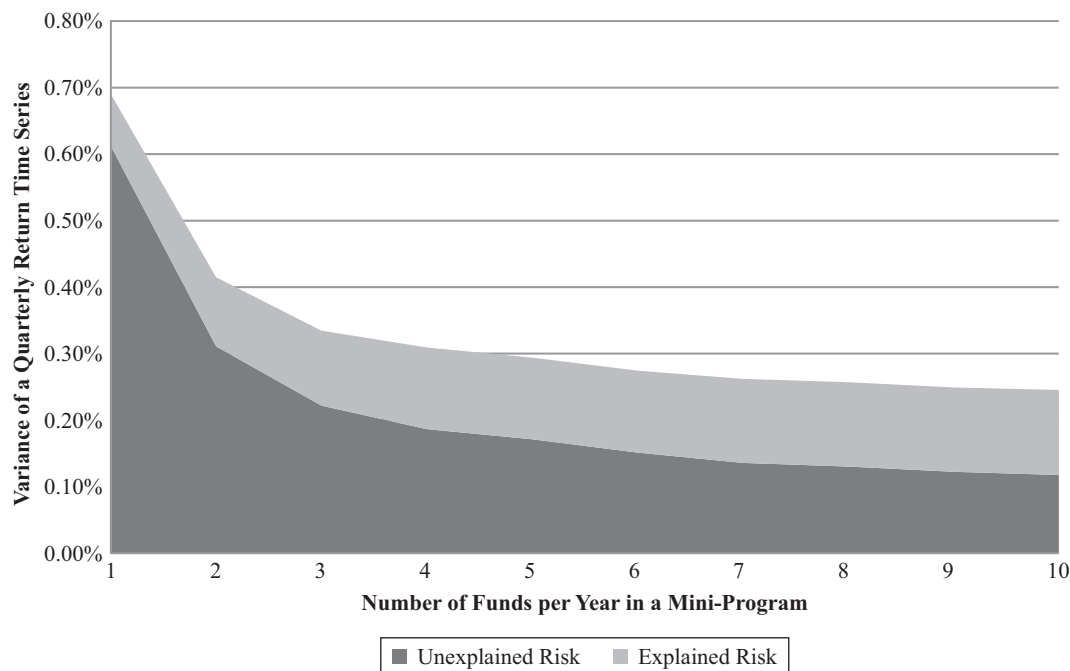
EXHIBIT 4

Factor Related vs. Idiosyncratic Risks: Data for 2000–2014

Panel A: Factor-Related vs. Idiosyncratic Risks for Venture Capital Mini-Programs



Panel B: Factor-Related vs. Idiosyncratic Risks for Buyout Mini-Programs



If anything, they decreased slightly after 2008, indicating a slight weakening of the link between private and public valuations.

Another noticeable difference between pre-2006 and post-2008 periods is the reduction of the smoothing

constant for buyouts. This is consistent with the notion that fair value accounting improved the quality of valuations and reduced smoothing.

The 2006–2008 period appears to be a real outlier supporting some (but not all) thoughts expressed

EXHIBIT 5

Analysis by Sample Period

	Buyout			Venture		
	Alpha	Theta	Beta	Alpha	Theta	Beta
Full Sample (2000–2014)	4.6%	0.39	0.53	1.0%	0.31	0.53
2000–2005	4.1%	0.41	0.53	–6.8%	0.18	0.50
2006–2008	9.5%	0.47	1.07	11.5%	0.14	0.78
2009–2014	6.2%	0.14	0.34	2.9%	0.51	0.45

in literature (see, e.g., Welch and Stubben 2018). This short but quite differentiated period incorporates two themes. On one hand, accounting changes in Europe and then in the United States prompted managers to adjust valuations from cost to market. In the case of European managers in 2006, this led to marking valuations up. Because equity markets were also up in 2006, this looked like an increase in public–private correlations but can hardly be described as such.

The effect of accounting changes on US valuation coincides entirely with the financial crisis, making the two very difficult to disentangle within our chosen framework. If one uses Welch and Stubben’s (2018) research on European managers as a guide, one may think that the two effects are similar in magnitude. In their study, which also indicated a sharp increase in private–public market correlation between the pre-2006 period and 2006–2008, about half of the increase came from the accounting-related markup. The other half of the increase is more puzzling. Welch and Stubben (2018) suggested that this increase reflects a permanent shift triggered by accounting changes unmasking a strong link between private and public valuations. In contrast, our study suggests that this spike was temporary in nature and that the private–public beta subsides after 2008. Such behavior stands to reason in our view: The whole equity asset class was repriced during these emotional times, with systemic risk factors dominating valuation changes across the board. Notably, private equity alpha came out as the highest during this time, quite possibly because the accounting-related markup came across—unfairly—as pure alpha in our type of analysis.

Given all these intricacies, the question of what we would suggest to use as private equity risk estimates going forward is not easy to answer. On reflection, our recommendation is to use the full sample results described in Exhibit 3. The sample there is the

longest, reducing estimation error, which is an important positive. Our analysis also suggests that accounting changes outside of the transition period of 2006–2008 had only a moderate effect on the risk model. Results of our study of idiosyncratic (residual) risks as a function of concentration of private equity mini-programs have been consistent throughout all periods; they are spelled out in Exhibit 3 as well. Finally, in the course of our study we found some evidence that private/public equity beta is likely to jump during times of financial stress. Private equity funds have a long life of 7–10 years or longer; it is likely that funds we purchase today will see another period of financial stress during their lifetime. Although equity betas shown in Exhibit 3 are higher than those observed post-crisis (Exhibit 4), using those higher estimates may still be prudent because they emphasize links between private and public markets that are stronger over the full business cycle than in the last few years, when market behavior was rather benign.

OPTIMAL ALLOCATION TO PRIVATE EQUITY WITHIN THE TOTAL PORTFOLIO

In the previous few sections we came up with a factor- and diversification-aware model for private equity strategies that demonstrated good explanatory power. We now turn to incorporating this model into an asset allocation framework. There is a great variety of such frameworks, including classical mean–variance optimization, a range of risk-focused approaches (see Da Silva et al. 2008; Lee 2011, Asness et al. 2012, and Rudin and Marr 2016 for additional literature), and other approaches incorporating illiquidity (e.g., Kinlaw, Kritzman, and Turkington 2013). We shall focus on the classical mean–variance framework, which is flexible, transparent, and most importantly extremely well known (with all the caveats).

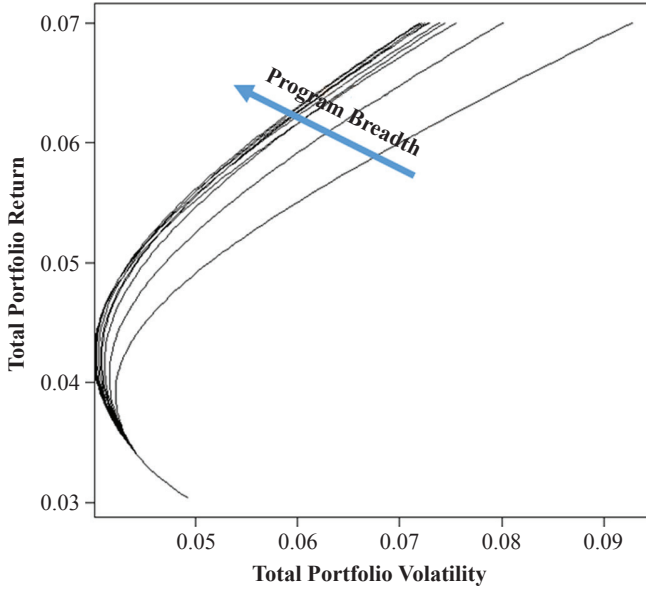
Using mean–variance as our framework, we now attempt to address two key questions: (1) What is the optimal allocation to private equity within the total portfolio, and (2) how should we evaluate trade-offs between the benefits of increased diversification and the costs of achieving it?

Needless to say, most asset allocation frameworks are highly sensitive to their inputs, and the results of our analysis should be viewed as illustrations of the concept instead of authoritative recommendations on specific allocations. Our attempt to study the interplay between diversification and optimal allocation to private equity

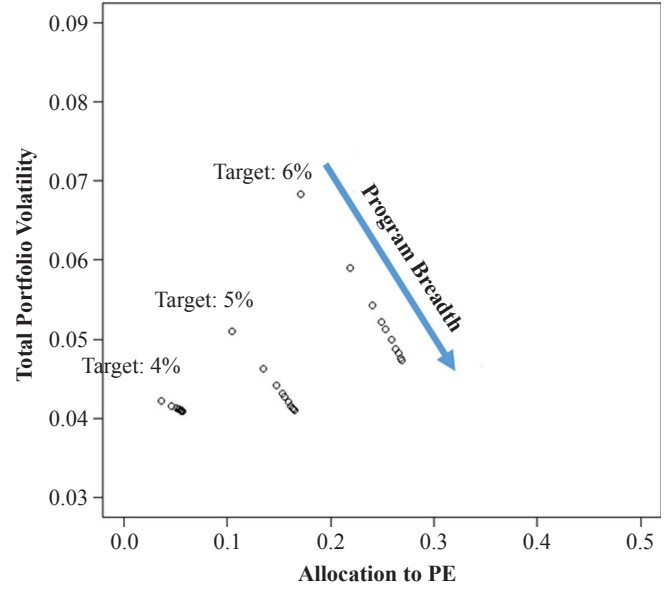
EXHIBIT 6

Efficient Frontier and Private Equity Allocation

Panel A: Efficient Frontier



Panel B: Recommended Private Equity Allocation



Notes: Efficient frontiers for portfolios including a private equity program with varying degree of diversification. Bonds are assumed to have expected annualized returns of 3% and volatility of 5%; public equities are assumed to have expected annualized returns of 7%, volatility of 13%, and correlation to bonds of -20%.

is the first of its kind. As such, we believe it is of great interest for institutional investors, despite the deliberate simplicity of our setup.

The portfolio we will study contains only three assets: bonds, public equities, and private equity (represented by buyout models from the previous section). For clarity, we chose the following form of mean–variance optimization:

$$\min(\mathbf{w}^T \mathbf{\Omega} \mathbf{w}) \quad (5)$$

Subject to:

$$\begin{aligned} \sum_i \mu_i w_i &= \mu^* \\ \sum_i w_i &= 1 \\ w_i &\geq 0 \end{aligned}$$

Here $\mathbf{w}$ is a vector of portfolio weights, $\mathbf{\Omega}$ a covariance matrix, and μ a vector of asset expected returns. Our constraints imply a long-only, fully invested portfolio that in aggregate targets an expected portfolio return of μ^* while minimizing volatility. Exhibit 6, Panel A illustrates efficient frontiers for a three-asset portfolio that

includes a private equity program of varying breadth. It also summarizes the assumptions for expected returns, volatilities, and correlations that we used for bonds and stocks. (Our private equity program assumptions are fully defined by Exhibit 3.)

Unsurprisingly, including a more diversified private equity program leads to portfolios that offer lower volatility for the same level of portfolio return. One can also show (see Exhibit 6, Panel B) that increasing diversification leads to having a higher optimal allocation to private equity for the same level of portfolio target return.

So far our analysis has ignored costs associated with increasing the breadth of a private equity portfolio. We now introduce these costs in the form of a linear correction to private equity expected returns:

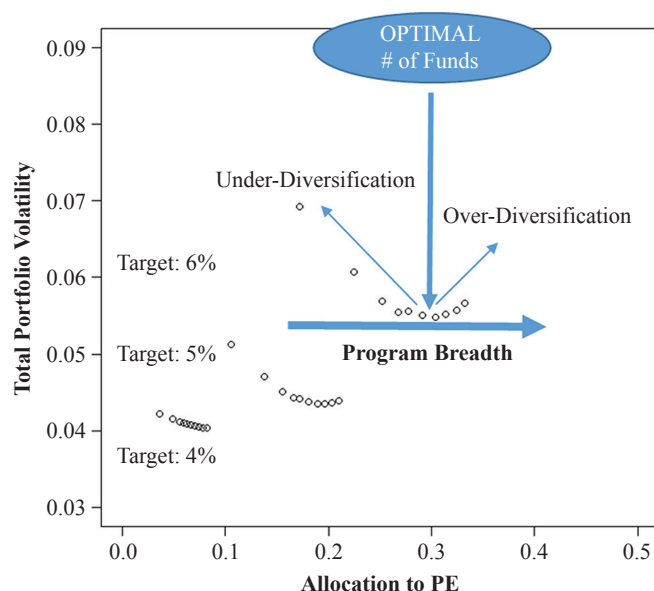
$$\mu_{PE}^{With\ Cost} = \mu_{PE}^{Before\ Cost} - N * c \quad (6)$$

Here c is a cost of adding a manager (per year) to the private equity program, and N is the number of managers.

Incorporating costs determined by Equation 6 into the private equity model of Exhibit 3 can lead to a

EXHIBIT 7

Optimal Number of Funds, Incorporating Costs



substantive change in the optimal number of funds, as shown in Exhibit 7.

Adding additional managers is no longer a cost-free exercise. Instead, there is a trade-off between the reduction in risk from increased diversification and the reduction in returns as a result of the additional costs. Because our portfolio optimization methodology aims to maximize returns per unit of risk, there is a point at which the breadth of the portfolio is optimal and adding more managers only leads to a deterioration in portfolio quality.

TAKEAWAYS AND CONCLUDING REMARKS

In this article we proposed an econometric model for private equity and tested the robustness of variations of this model on a specially created fund dataset. Through this process we uncovered several noteworthy takeaways.

First, we were able to formally compare models of varying complexity and demonstrate that an increasing number of lags is actually detrimental to the model's forecasting power. Instead, settling on a single-lag model proved to be the most robust approach in almost all cases. We believe this result (and our method of obtaining it) is useful to both academics and practitioners involved

in managing portfolios that include private equity. It may also augment (otherwise) very insightful work by Pedersen, Page, and He (2014), who used a five-lag model for private equity.

Second, because we were able to calibrate our risk estimation model on a dataset with tightly controlled diversification properties, we were able to distill the amount of idiosyncratic error as a function of portfolio concentration. This finding can help institutional investors estimate the impact of the narrowness (or breadth) of their program on expected returns and their variability.

Finally—and we believe most importantly—we demonstrated that the amount of equity exposure (beta) that one ought to assign to private equity is material but significantly lower than suggested by others. This is a very consequential result that may affect how institutional investors allocate to private equities within their overall portfolios; it warrants additional comments.

Part of the explanation for why our estimate of private equity beta is lower than that from other authors is technical. Some authors, we believe, relied on too many lags, causing their models to overfit and overstate the beta (see Exhibit 1 as an illustration). Another reason why we obtained lower levels of beta is our use of an equal commitment methodology for constructing our private equity mini-programs. This methodology is representative of the way most real-life private equity programs are constructed but is in contrast to how typical private equity indexes are constructed; the latter are usually capitalization weighted and dominated by a few large funds, with those funds tending to have a higher level of leverage and hence beta.

That said, the main reason why private equity beta is so low is deeply fundamental. Ever since Robert Shiller came up with the concept of *excess volatility* (see Shiller 1981), it has been understood that public equity gyrations are driven partly by earnings expectations and partly by psychological phenomena (e.g., anxiety, flows, momentum). It is self-evident that although private equity valuations processes are designed to fully reflect earnings expectations of underlying companies, they are much less exposed to excess volatility than their public counterparts. This insight helps reconcile two observations seemingly at odds with each other: our findings of private equity having moderate to low equity beta over the short term (quarter by quarter) and earlier findings by L'Her et al. (2016) and Phalippou (2014),

who witnessed “no significant outperformance of buyout fund investments versus their public market equivalents” over the very long term (10+ years). In the short term, excess volatility that dominates public markets does not enter the private equity valuations in a material way, which explains why private equity betas and volatility are lower than comparable public benchmarks. Conversely, over the very long term, excess volatility washes away and earnings are the only thing that truly matters, making long-term results of public and private equity investing similar to each other.

REFERENCES

- Anson, M. 2002. “Managed Pricing and the Rule of Conservatism in Private Equity Portfolios.” *The Journal of Private Equity* 5, no. 2 (Spring): 18–30.
- Asness, C., A. Frazzini, and L. Pederson. 2012. “Leverage Aversion and Risk Parity.” *Financial Analysts Journal* 68 (1): 47–59.
- Brandt, M. W. “Portfolio Choice Problems.” In *Handbook of Financial Econometrics, Volume 1: Tools and Techniques*, edited by Y. At-Sahalia and L. P. Hansen, pp. 269–336. Amsterdam: North-Holland, 2010.
- Brown, G. W., O. R. Gredil, and S. N. Kaplan. 2018. “Do Private Equity Funds Manipulate Reported Returns?” *Journal of Financial Economics* 132 (2): 267–297.
- Conner, A. 2003. “Asset Allocation Effects of Adjusting Alternative Assets for Stale Pricing.” *The Journal of Alternative Investments* 6, no. 3 (Winter): 42–52.
- Crain, N., and K. Law. 2017. “The Bright Side of Fair Value Accounting: Evidence from Private Company Valuation.” Working paper.
- Da Silva, A., W. Lee, and B. Pornrojngkool. “Risk Budgeting.” *Handbook of Finance: Investment Management and Financial Management*, Vol. II, Chapter 21, pp. 195–220. Hoboken, NJ: John Wiley & Sons, 2008.
- Easton, P. D., S. Larocque, and J. S. Stevens. 2018. “Private Equity Valuation Before and After ASC 820.” Working paper.
- Geltner, D. 1991. “Smoothing in Appraisal-Based Returns.” *The Journal of Real Estate Finance and Economics* 4 (3): 327–345.
- Jenkinson, T., W. R. Landsman, B. Rountree, and K. Soonawalla. 2016. “Private Equity Net Asset Values and Future Cash Flows.” Working paper.
- Kinlaw, W., M. Kritzman, and D. Turkington. 2013. “Liquidity and Portfolio Choice: A Unified Approach.” *The Journal of Portfolio Management* 39 (2): 19–27.
- Lee, W. 2011. “Risk-Based Asset Allocation: A New Answer to an Old Question?” *The Journal of Portfolio Management* 37 (4): 11–28.
- L’Her, J. F., R. Stoyanova, K. Shaw, W. Scott, and C. Lai. 2016. “A Bottom-Up Approach to the Risk-Adjusted Performance of the Buyout Fund Market.” *Financial Analysts Journal* 72 (4): 36–48.
- Liew, V. K. S. 2004. “What Lag Selection Criteria Should We Employ?” *Economics Bulletin* 33 (3): 1–9.
- Pedersen, N., S. Page, and F. He. 2014. “Asset Allocation: Risk Models for Alternative Investments.” *Financial Analysts Journal* 70, no. 3 (May/June): 34–45.
- Phalippou, L. 2014. “Performance of Buyout Funds Revisited.” *The Review of Finance* 18, no. 1 (January): 189–218.
- Schiller, R. J. 1981. “Do Stock Prices Move Too Much To Be Justified by Subsequent Changes in Dividends?” *The American Economic Review* 71, no. 3 (June): 421–435.
- Schmidt, D. “Private Equity-, Stock- and Mixed Asset-Portfolios: A Bootstrap Approach to Determine Performance Characteristics, Diversification Benefits and Optimal Portfolio Allocations.” CFS Working Paper Series 2004/12, Center for Financial Studies.
- Stefek, D., and R. Suryanarayanan. 2012. “Private and Public Real Estate: What Is the Link?” *The Journal of Alternative Investments* 14 (3): 66–75.
- Rudin, A., W. M. Marr. 2016. “Investor Views, Drawdown-Based Risk Parity, and Hedge Fund Portfolio Construction.” *The Journal of Alternative Investments* 19 (2): 63–69.
- Welch, K. T., and S. Stubben. 2018. “Private Equity’s Diversification Illusion: Evidence From Fair Value Accounting.” Working paper.

To order reprints of this article, please contact David Rowe at d.rowe@pageantmedia.com or 646-891-2157.

ADDITIONAL READING

Managed Pricing and the Rule of Conservatism in Private Equity Portfolios

MARK J.P. ANSON

The Journal of Private Equity

<https://jpe.pm-research.com/content/5/2/18>

ABSTRACT: All private equity funds face the difficult issue of placing a value on the worth of their private equities. The rule of conservatism states that portfolios should be marked down quickly and up slowly, and that a triggering event, such as a new round of investment, should provide an objective, market-driven valuation point. This article determines the performance in non-market equity pricing for three classes of private equity (venture capital, leveraged buyouts, and mezzanine debt) against the U.S. Treasury bill, Nasdaq, Russell 1000, and Russell 2000 benchmark rates of return.

Asset Allocation Effects of Adjusting Alternative Assets for Stale Pricing

ANDREW CONNER

The Journal of Alternative Investments

<https://jai.pm-research.com/content/6/3/42>

ABSTRACT: As institutional interest in alternative asset classes such as private equity and hedge funds has grown, plan sponsors are confronted with difficulties applying traditional asset allocation models. Estimating the risk and correlation parameters of these asset classes is often less than straightforward. One culprit is the infrequent and appraisal-based pricing necessitated by investing in illiquid securities. This “stale pricing” can reduce the perception of volatility. In this study we apply a methodology for estimating true volatility and correlation in the presence of stale pricing to alternative investments.

We also measure the impact of adjusting for stale pricing on asset allocation and the diversification benefits of private equity and hedge fund investments. We conclude that much of the perceived diversification benefits associated with allocating to private equities and hedge funds is attributable to stale pricing and thus illusory. Nonetheless, we observe that efficient portfolios still contain substantial allocations to alternative investments after adjusting for the effects of stale pricing.

Liquidity and Portfolio Choice: A Unified Approach

WILL KINLAW, MARK KRITZMAN, AND DAVID TURKINGTON

The Journal of Portfolio Management

<https://jpm.pm-research.com/content/39/2/19>

ABSTRACT: The authors propose a simple analytical construct for incorporating liquidity into portfolio choice. In cases where investors deploy liquidity to raise a portfolio's expected utility beyond its original expected utility, the authors attach a shadow asset to tradable assets. In cases where investors deploy liquidity to prevent a portfolio's expected utility from falling, the authors attach a shadow liability to assets that are not tradable. This construct lets investors determine the optimal allocation to illiquid assets. Alternatively, investors can use this construct to estimate the premium an illiquid asset requires, or the degree to which they must benefit from liquidity in order to justify forgoing investment in illiquid assets. This approach improves on other methods of incorporating liquidity into portfolio choice in four fundamental ways. First, it mirrors what actually occurs within a portfolio. Second, it maps units of liquidity onto units of expected return and risk, so that investors can analyze liquidity within the same context as other portfolio decisions. Third, it distinguishes absolute illiquidity from partial illiquidity and lets investors address these attributes within a single, unifying framework. Fourth, it recognizes that liquidity serves not only to meet demands for capital, but to exploit opportunities as well, thus revealing that investors bear an illiquidity cost to the extent that any fraction of a portfolio is immobile.

Reproduced with permission of copyright owner. Further reproduction
prohibited without permission.